Microprocessor System Design Considerations

SoC Technology Roadmap and Trend

1. A portable Audio/Video Media player is designed based on the specification given below.
    - Li-Polymer Battery.
    - Audio Playtime 24hr, Video Play-time 10hr, Voice Recording 10hr.
    - USB3 slave connectivity
    - Standby time 7 days.
    - 1920x1080 LCD with Capacitive Touch Screen (SPI interface)
    - 8 user input buttons
    - System wakeup from standby via any button and SPI activity within 10μs
    - Software
        - Main system overhead: 20MIPS (Million Instruction per second)
        - MP3 decoding: 15 MIPS
        - MP4 decoding: 50 MIPS
        - Voice recording: 5 MIPS
        - Total program code size: 240KByte
        - Total program data size: 128Kbyte
        - Total data buffer size: 256MByte
    - Processor power consumption
        - System Overhead: 5mA/MIPS
        - Active (Audio/Voice functions): 5mA/MIPS
        - Active (Audio+Video functions): 10mA/MIPS
        - Standby: 3mA
    - Processor internal memory
        - 256 KByte SRAM, 512 KByte Flash

    (a) What external storage and system memory is suitable for this product? Or are they required?

    (b) Does the processor require any special internal peripheral to interface to the memories listed above?

    (c) Calculate the power consumption when performing the following function and calculate the battery capacity required (in mAh) to meet the requirement specified. MP4 includes audio and video. Assume system overhead is the same for all activities. Ignore power consumed by other components on the player.
        - Audio Decode
        - MP4 Decode
        - Voice Record

- Standby (System Sleeping)

(d) What could be done to reduce the battery capacity but still able to meet the various product function requirement?

(e) What could be done if we want to increase the standby time?

(f) Audio codec typically requires a special clock frequency that is a multiple of the sampling frequency. Assuming that the sampling frequency, fs=44.1Khz, one of the common clock frequencies used is 256fs. How can this frequency be generated for this system? External Crystal used by the Processor has a frequency of 48Mhz.

(g) Name one important reason why the processor needs a 48MHz crystal.

2. With reference to the information derived in Q1, some questions below

   - Based on the very fast wakeup time requirement of 10μs, should a crystal be used during standby or an on-chip oscillator?
   - What other points should be considered where pin connection and termination are concerned?

3. Further to the system in Q1, what are the power supply design considerations given the following?
   - The processor selected has separate voltage rails for its core and I/O.
   - The system includes mixed signal components such as audio codec, video codec, ADC etc.
   - Any other additional power management considerations.

4. What is value proposition of the heterogeneous processing architecture that ARM proposed? How does ARM's big.LITTLE™ processor architecture managed to fulfill the these value proposition?